

vol. in pov. vrtenine

$$\vec{r}(t) = (x(t), y(t))$$

$$V = \int_a^b \pi y^2 dx = \pi \int_a^b y(t)^2 x'(t) dt$$

$$P = 2\pi \int_a^b y \sqrt{x'^2 + y'^2} dt$$

①. $\vec{r}(t) = (\sin^2 t, t + \sin t \cos t) \quad t \in [0, \frac{\pi}{2}]$

$P, V = ?$

zau. okoli x osi

$$(\sin^2 t)' = 2 \cos t \sin t$$

$$y'(t) = 1 - \sin^2 t + \cos^2 t = 2 \cos^2 t$$

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{2}} (t + \sin t \cos t)^2 \cdot 2 \cos t \sin t dt = 2\pi \int_0^{\frac{\pi}{2}} (t^2 + 2t \sin t \cos t + \sin^2 t \cos^2 t) \cos t \sin t dt = \\ &= 2\pi \underbrace{\int_0^{\frac{\pi}{2}} t^2 \sin t \cos t dt}_1 + 2\pi \underbrace{\int_0^{\frac{\pi}{2}} 2t \sin^2 t \cos^2 t dt}_2 + 2\pi \underbrace{\int_0^{\frac{\pi}{2}} \sin^3 t \cos^3 t dt}_3 \end{aligned}$$

③. $\sin t = u$
 $\cos t dt = du$

$$2\pi \int_{u(0)}^{u(\frac{\pi}{2})} u^3 \cos^2 t du = 2\pi \int_0^1 u^3 du - 2\pi \int_0^1 u^5 du = 2\pi \left(\frac{u^4}{4} - \frac{u^6}{6} \right) \Big|_0^1 = 2\pi \left(\frac{1}{4} - \frac{1}{6} \right) =$$

$$= 2\pi \left(\frac{3-2}{12} \right) = \underline{\underline{\frac{\pi}{6}}}$$

②. $4\pi \int_0^{\frac{\pi}{2}} t \sin^2 t \cos^2 t dt = 4\pi \int_0^{\frac{\pi}{2}} t \sin^2 t dt - 4\pi \int_0^{\frac{\pi}{2}} t \sin^4 t dt$

$$P = \int_0^{\frac{\pi}{2}} 2\pi(t + \sin t \cos t) \sqrt{4\cos^2 t \sin^2 t + 4\cos^4 t} dt =$$

$$= 4\pi \left[\int_0^{\frac{\pi}{2}} t \cos t dt + \int_0^{\frac{\pi}{2}} \sin t \cos^3 t dt \right] = 4\pi \left(\frac{\pi}{2} - \frac{2}{3} \right)$$

1. Izračunaj Tayl. polinoma

a) reda 3 funkcije $f(x) = xe^{-x^2}$ u okolini $a=0$

$$f'(x) = -2x^2 e^{-x^2} + e^{-x^2} = e^{-x^2}(1 - 2x^2)$$

$$f''(x) = \dots$$

3. Oцени napako aproksimacije

$$\operatorname{arctg} x \approx x \quad \text{za } |x| < \frac{1}{2}$$

$$R_n(x) = \frac{f^{(n+1)}(t)}{(n+1)!} (x-a)^{n+1}$$

$$t \in (a, x) \vee (x, a)$$

$$\text{na int. } |x-a| < \delta : |R_n| \leq \max_{|t-a| < \delta} |f^{(n+1)}(t)| \frac{\delta^{n+1}}{(n+1)!}$$

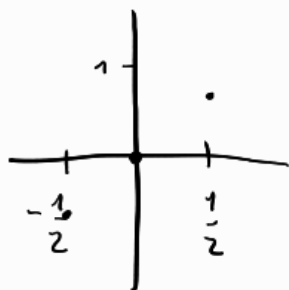
$$T_1 f(x, 0) = x$$

$$f'(x) = \frac{1}{1+x^2}$$

$$|R_1| \leq \max_{|t| < \frac{1}{2}} \left| -\frac{2x}{(1+x^2)^2} \right| \frac{\frac{1}{2}}{2} < \frac{1}{8}$$

$$f''(x) = -\frac{1}{(1+x^2)^2} \cdot 2x$$

$$g(x) = \frac{2x}{(1+x^2)^2}$$



Če bi vzeli, da je $\operatorname{arctg} x \approx x$
Taj. pol. 2. st., bi dobili ostaneke
manjši

4. Razvij ok. $a=0$

a) $f(x) = \operatorname{sh} x \quad f(0) = 0$

$$f'(x) = \operatorname{ch} x \quad f'(0) = 1$$

$$f''(x) = \operatorname{sh} x$$

$$T_{2n+1} f(x, 0) = \sum_{j=0}^n \frac{x^{2j+1}}{(2j+1)!} \approx \operatorname{sh} x$$

$$f(x) = \operatorname{ch} x \quad f(0) = 1$$

$$f'(x) = \operatorname{sh} x \quad f'(0) = 0$$

$$f''(x) = \operatorname{ch} x$$

$$T_{2n} f(x, 0) = \sum_{j=0}^n \frac{x^{2j}}{(2j)!} \approx \operatorname{ch} x$$

$$\operatorname{ch}(iz) = \cos z \quad ; z \in \mathbb{C}$$

$$\operatorname{sh}(iz) = i \sin z$$

$$b) f(x) = \frac{1}{(1-x)^k}; k \in \mathbb{N}$$

$$= 1 - \binom{-k}{1}x + \binom{-k}{2}x^2 - \dots = \sum_{h=0}^{\infty} \frac{(-k)(-k-1)\dots(-k-h+1)}{h!} (-x)^h =$$

$$= \sum_{h=0}^{\infty} \frac{k(k+1)\dots(k+h-1)}{h!} (x^h) = \sum_{h=0}^{\infty} \binom{k+h-1}{h} x^h$$

$$c) f(x) = \frac{1}{x^2-3x+2} = \frac{1}{(2-x)(1-x)} = \frac{1}{1-x} - \frac{1}{2-x} = \frac{1}{1-x} - \frac{1}{2} \frac{1}{1-\frac{x}{2}} =$$

$$= \sum_{h=0}^{\infty} \binom{-1}{h} (-x)^h - \frac{1}{2} \sum_{h=0}^{\infty} \binom{-1}{h} \left(-\frac{x}{2}\right)^h =$$

$$\binom{-1}{h} = \frac{(-1)(-1-1)\dots(-1-h+1)}{h!} = \frac{(-1)^h h!}{h!}$$

$$\left\{ \begin{array}{l} \frac{A}{2-x} + \frac{B}{1-x} = \frac{1}{(2-x)(1-x)} \\ 4-Ax+2B-Bx=0 \\ -A-B=0 \quad A=-B \\ A+2B=1 \end{array} \right.$$

$$= \sum_{h=0}^{\infty} x^h - \sum_{h=0}^{\infty} x^h \frac{1}{2^{h+1}} = \sum_{h=0}^{\infty} \left(1 - \frac{1}{2^{h+1}}\right) x^h$$

$$\boxed{\begin{array}{l} B=1 \\ A=-1 \end{array}}$$

$$d) f(x) = \frac{\ln(x+1)}{x+1} = \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\right) \left(1 - x + x^2 - x^3 + \dots\right) =$$

$$a_0 = 0$$

$$a_1 = 1$$

$$a_2 = -1 - \frac{1}{2}$$

$$a_3 = 1 + \frac{1}{2} + \frac{1}{3}$$

$$a_n = (-1)^{n+1} \sum_{k=1}^n \frac{1}{k}$$

$$= \sum_{h=1}^{\infty} (-1)^{h+1} \cdot x^h \cdot \sum_{k=1}^h \frac{1}{k}$$

4. Razvij funkcijo

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-\frac{t^2}{2}} dt$$

$$e^{-\frac{t^2}{2}} = 1 + \left(-\frac{t^2}{2}\right) + \frac{\left(-\frac{t^2}{2}\right)^2}{2} + \frac{\left(-\frac{t^2}{2}\right)^3}{3!} + \dots$$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \left(x - \frac{x^3}{6} + \frac{x^5}{40} - \frac{x^7}{336} + \dots \right)$$

5. Izračunaj lim s Taylorjem

a) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1 - \frac{1}{2}x^2 + \frac{1}{8}x^4}{x^6} =$

$$\sqrt{1+x^2} = (1+x^2)^{\frac{1}{2}} = 1 + \binom{\frac{1}{2}}{1}x^2 + \binom{\frac{1}{2}}{2}x^4 + \binom{\frac{1}{2}}{3}x^6 = 1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \frac{1}{16}x^6$$

$$\binom{\frac{1}{2}}{1} = \frac{\frac{1}{2}}{1} = \frac{1}{2} \quad \binom{\frac{1}{2}}{3} = \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{6} = \frac{3}{6 \cdot 8} = \frac{1}{16}$$

$$\binom{\frac{1}{2}}{2} = \frac{\frac{1}{2} \cdot \left(-\frac{1}{2}\right)}{2} = -\frac{1}{8}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{16}x^6}{x^6} = \underline{\underline{\frac{1}{16}}}$$

b) $\lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{1 + x^2 + \frac{x^4}{2} + \frac{x^6}{3!} + \dots - 1}{1 - \left(1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots\right)} = \lim_{x \rightarrow 0} \frac{\left(x^2 + \frac{x^4}{2} + \frac{x^6}{3!} + \dots\right) : x^2}{\left(\frac{x^2}{2} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots\right) : x^2} =$

$$= \lim_{x \rightarrow 0} \frac{1 + \frac{x^2}{2} + \dots}{\frac{1}{2} - \frac{x^2}{4!} + \dots} = 2$$